Queueing disciplines and switches

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CS 356

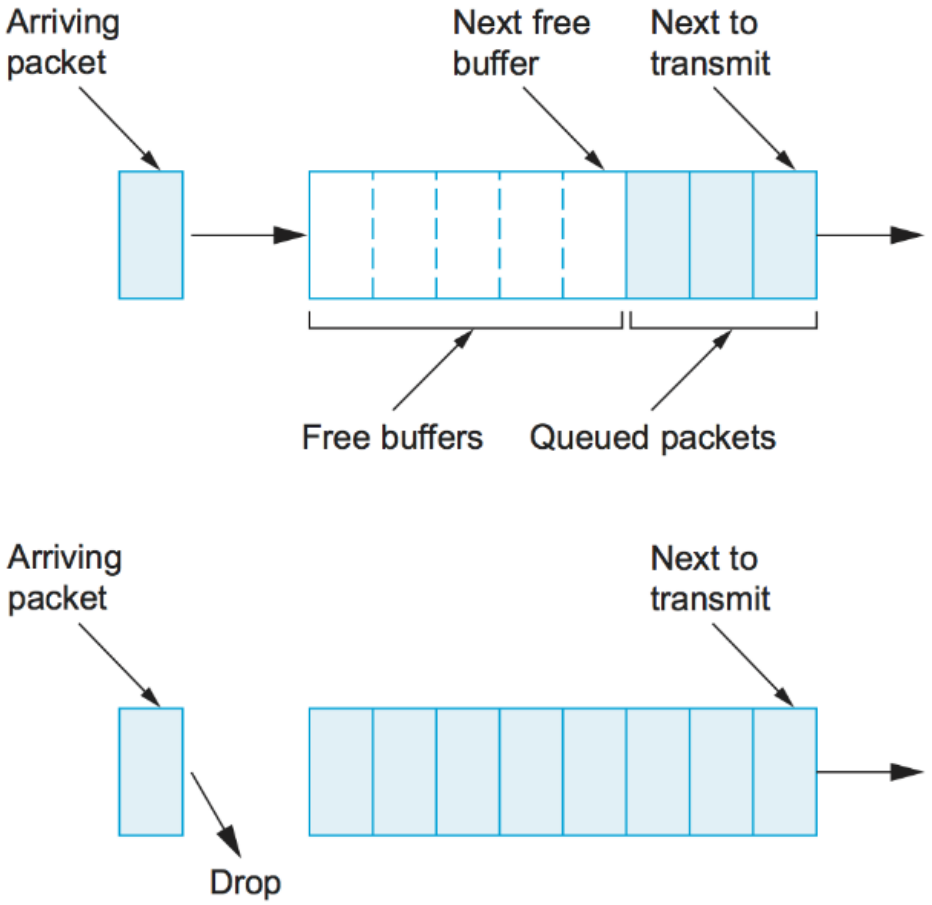
Some figures are borrowed from Nick McKeown’s slides

Theme of today’s class

- Subtle differences in design decisions lead to strange outcomes

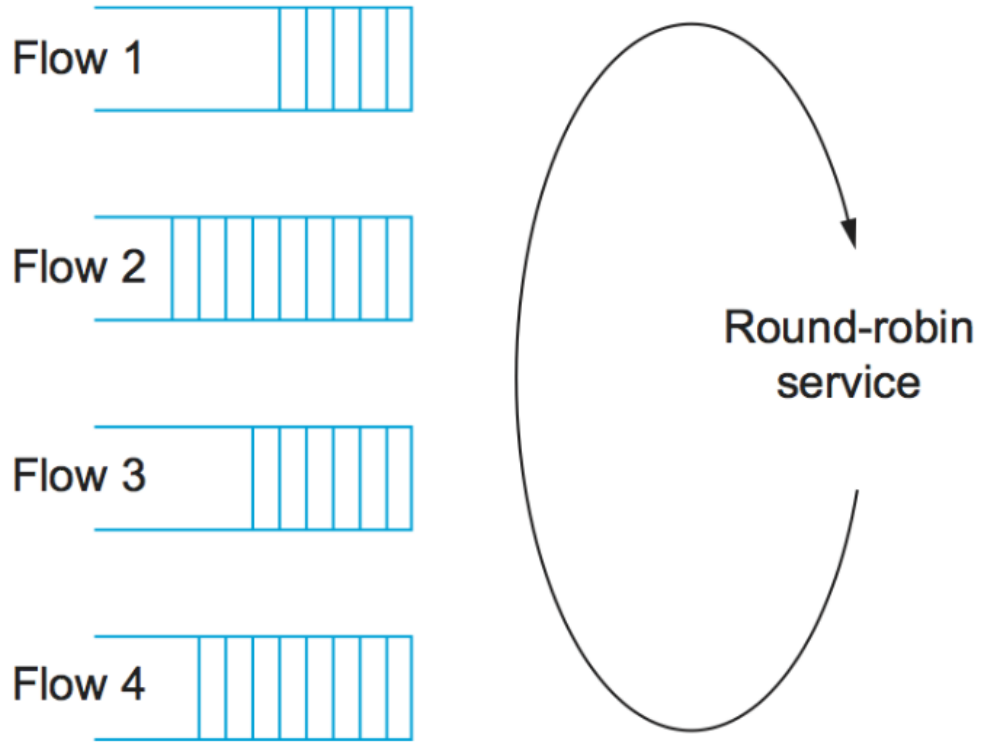
Queuing discipline: FIFO

- This is what we had assumed when we discussed AIMD congestion control
- All packets arriving at a port are enqueued here
- Advantage: simple to implement
- Disadvantage 1: Flows that send more get more bandwidth. They can go “rogue”, blast packets into the network and get rewarded for it (e.g. by modifying TCP congestion control)
- Disadvantage 2: Some flows want throughput. They want to maintain a larger queue so that if the link rate suddenly increases, there are packets to transmit. Others prefer lower latency, but their packets will be stuck behind the throughput-sensitive flows’ packets



Queuing discipline: Fair queuing

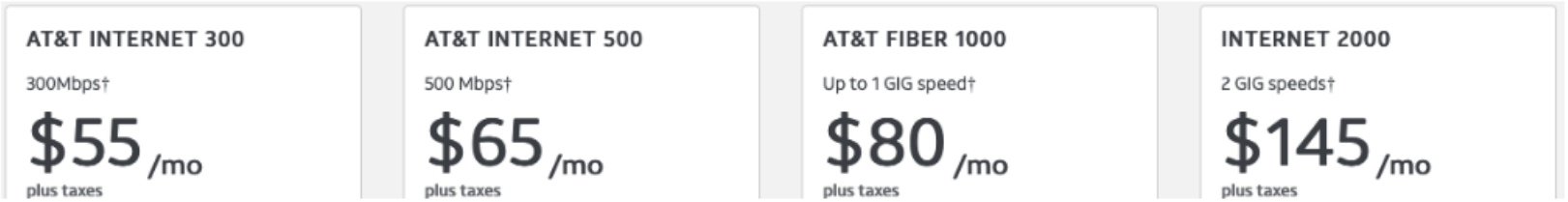
- Input queues discussed earlier are split further into multiple queues.
- Packets are classified into “flows” in someway and put into separate queues
  - E.g. using src/dst IP and port and the protocolnumber. This is commonly called a “5 tuple”
- Packets are dequeued so that each queue gets the same number of bytes per second
- Advantage: prevents rogue flows (Q. does it?)
- Disadvantage: much more complex. In particular, hardware is bad at maintaining a variable number of queues since thenumber of packets is not known a priori



Queuing discipline: artificial bottlenecks

- Sometimes routers are explicitly instructed to forward at a lower speed than what they are capable
- This is extremely simple to implement and ensures the congestion stays near the edges of the internet.
- Mechanisms like fair queuing are nearly impossible to implement near the core where millions of flows may go through the same link

Commercial artificial bottleneck



- This usually just means that there is some place between you and the network that is bottlenecked at the advertised speed.
- It caps the maximum, but says very little about what speed you will actually get.
- When you change your plan, nobody is going out to install a new router or wire. It is just a software change that asks the router to change how much it is throttling your packets.
- ISP contracts with commercial entities (e.g. UT Austin) are similar, except with bigger numbers

Routers

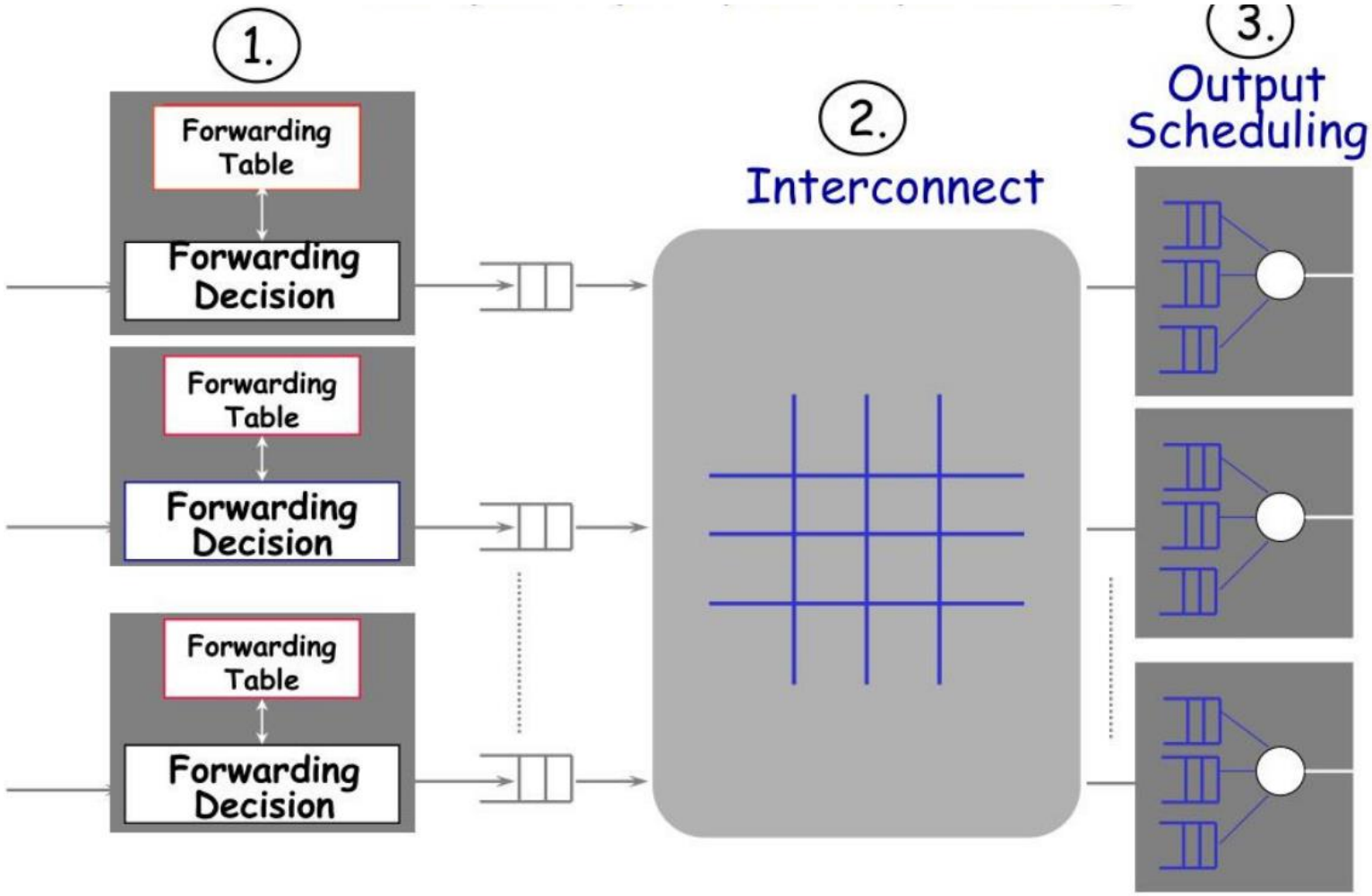


- A router is more than a single queue. It takes packets from one port (pictured), reads its destination address and forwards it to the appropriate port
- How does it know the destination port? By looking up the destination address in the routing table
- How do you implement this efficiently in hardware?

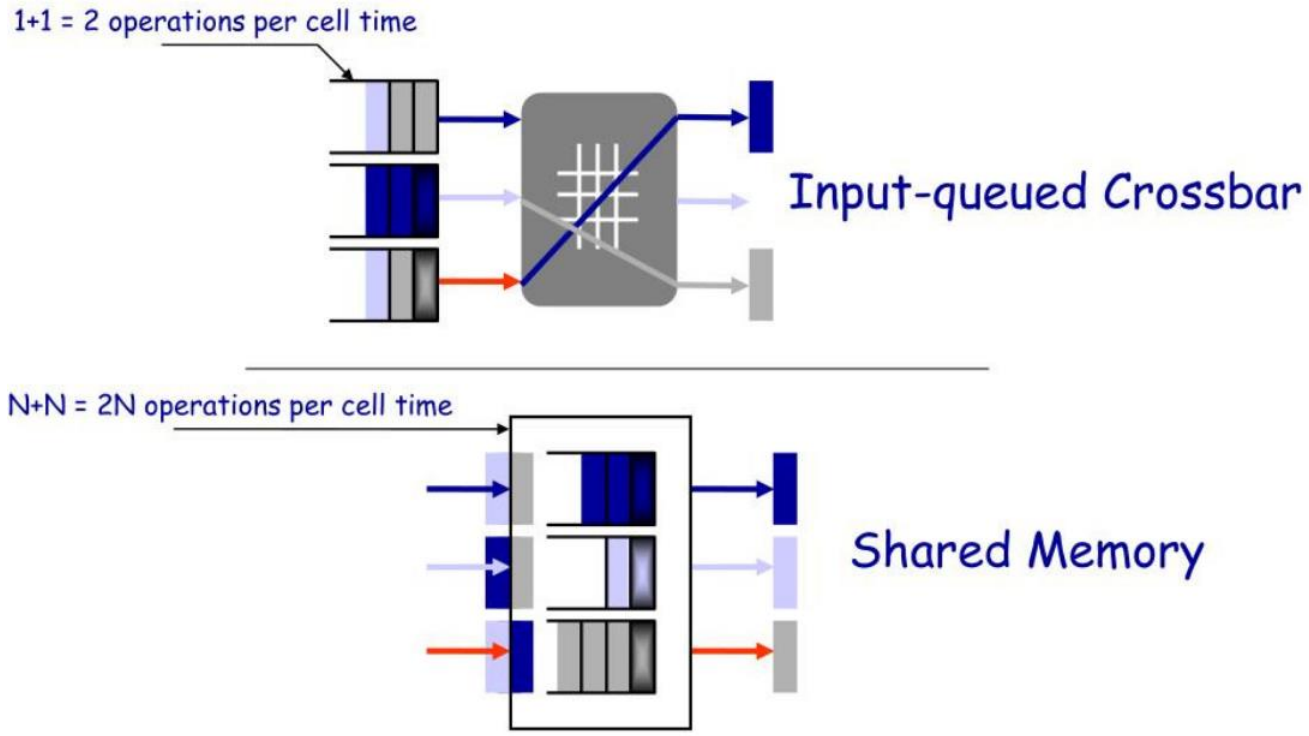
How do you switch packets?

1. When a packet comes, you decide where to forward it
2. You send it to the output port somehow
3. You decide the order in which to send packets at the output port (Q. Why can't you do this at the input?)

Note: we treat input ports as distinct from output ports. Every physical port has an input and output port



Shared memory architecture vs input-queued crossbar

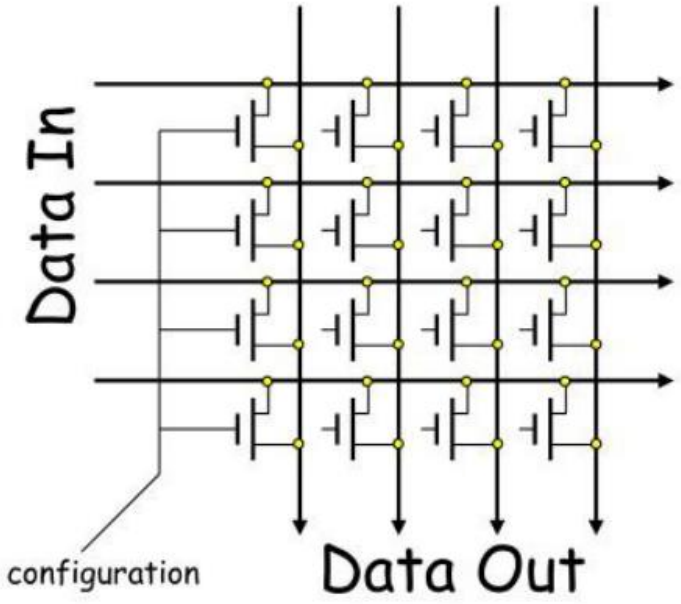


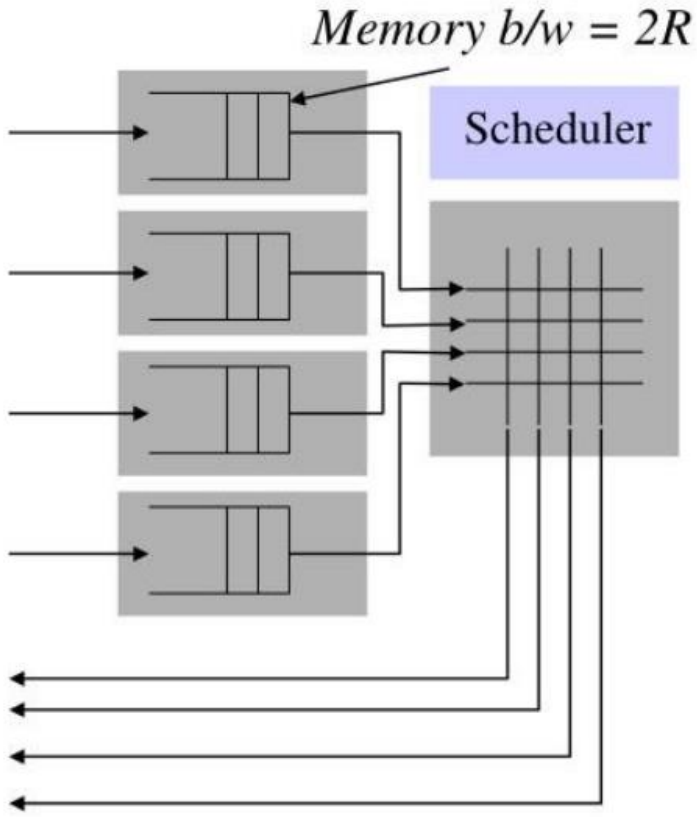
Input queued: Ports act independently except for the crossbar which takes packets from the input to the output side. Fragments memory

Shared memory: Input ports write to a datastructure in a shared memory that output ports read. Requires too much memory bandwidth!

High speed switches today use crossbars

Crossbars

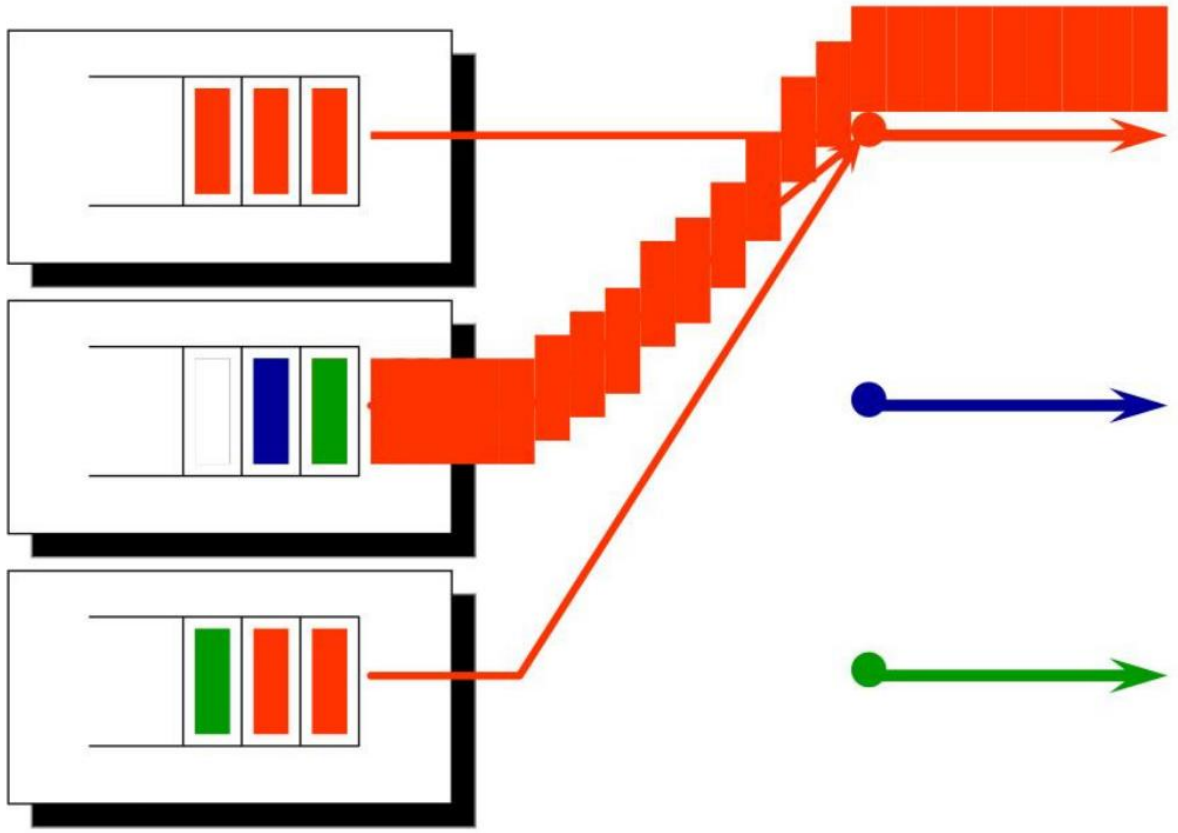




- The crossbar (left) selects which input port is connected to which output port. Input (output) ports can be connected to at most one output (input) port
- You can imagine this as a bipartite graph (shown on board)

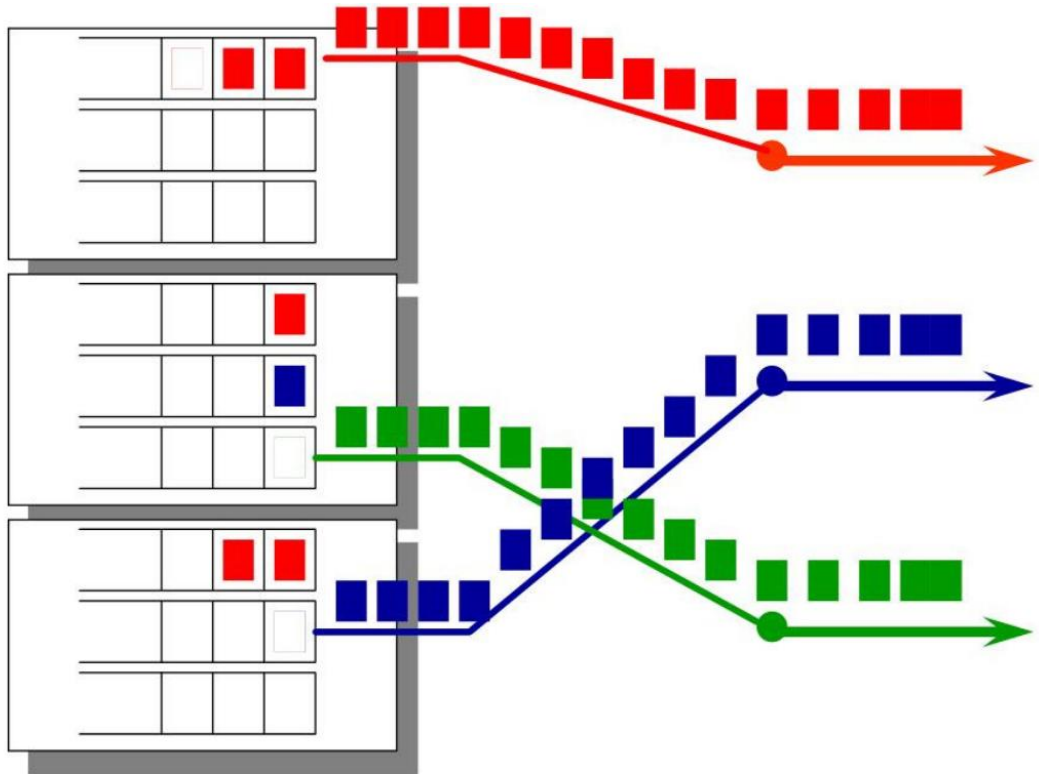
Head of line blocking

- Greed and blue packets can be transmitted in parallel with the red ones, but are blocked by the red packets



Head of line blocking solved with VOQs

- Each input queue was split into a separate “Virtual Output Queue (VOQ)”, one for each output
- This solves the head-of-line blocking problem



Questions to think about

- Suppose the input ports, output ports and crossbar operate at the same speed, where will queuing occur?
- In such switches, fair queuing is often done at the input queue. What does this mean for fairness?
- If you want queues to only exist in the output ports, what design constraints must you obey?